Work-let Name: On device Topic modelling for small text data

Worklet Details

1. Worklet ID: 25PI05TU
2. College Name: Thapar Institute of Engineering & Technology

KPIs achieved till now

- Completed data preparation pipeline using RAKE for automatic distant supervision (no manual labeling required).
- Successfully fine-tuned MobileBERT architecture framing topic-modelling as Token Classification.
- Achieved expected evaluation metrics: F1-Score ~0.83, Precision ~0.85, Recall ~0.80.
- Successfully quantized the model for mobile deployment (size reduced to ~7.0 MB).

Any Challenges/ Issues faced

- **Task Framing**: Traditional topic modelling (like LDA) or generative models are too heavy for mobile. We overcame this by framing it as a Token Classification task.
- **Token Alignment**: Aligning word-level RAKE pseudo-labels to WordPiece sub-tokens in MobileBERT required careful masking for accurate PyTorch loss calculation.
- **TFLite Conversion constraints**: Exporting deep learning models to TFLite often breaks; resolved this efficiently by using HuggingFace's from_pt=True directly into a TensorFlow graph before applying Dynamic Range Quantization.

Next Steps

- Develop a minimal Android UI to test the model on actual hardware.
- Integrate the .tflite model using Android ML Kit / TensorFlow Lite interpreter.
- Benchmark on-device memory performance and inference latency.
- Begin drafting the technical paper detailing the research and implementation process.

Key Achievements/ Outcome till now

- Fully functional end-to-end Python pipeline (from download_data.py to predict.py).
- Proved that a 7MB quantized MobileBERT model can confidently extract accurate contextual topics from short Android notifications/SMS completely offline.

Date: 29th April, 2026

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Resource Requirement Identified with Status

- Compute Resources for Training: GPU environment for fine-tuning MobileBERT (Status: Secured/Completed).
- Testing Hardware: Android device for testing on-device inference latency and real-world memory profiling (Status: Identified / In Progress).

Github Repository Link and PRISM Portal Registration Status

Github Repository Link - https://github.com/priyanshigoyal2801/25PI05TU
Portal Registration Status - Completed

Any Anticipated Break (Due to exams etc)

End Semester Examinations - 11th May - 21st May
Internship Period - June & July

Data Collection Information

- **Source Type:** Open Source
- **Dataset Name:** UCI SMS Spam Collection Dataset (~5,574 English SMS messages).
- **Details**: Data is automatically downloaded and parsed by our download_data.py script. We apply RAKE algorithm for unsupervised pseudo-labeling.
- Dataset Link: https://archive.ics.uci.edu/ml/datasets/sms+spam+collection

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Professor Comments (To be filled in and presented by professor (Either)

Status and future plan
Latest Literature Review
Expert Insights
Any other comment

COMPUTATION RESOURCE Details:

Local PC (Laptop with dedicated GPU). Total compute hours: ~0.5 to 1 hour for full end-to-end dataset preprocessing, model fine-tuning (5 epochs), and TFLite conversion.

Date: xxxxxxx